

1) List all 6 phases of SDLC in order with one line about each.

Ans:

1. **Requirement Analysis** – Gather and analyze the needs and requirements of the users and stakeholders.
2. **System Design** – Create the system architecture, database design, and technical specifications.
3. **Implementation (Coding)** – Developers write the program code according to the design.
4. **Testing** – Test the software to find and fix bugs and ensure it works correctly.
5. **Deployment** – Release the software to users or the production environment.
6. **Maintenance** – Provide updates, fix issues, and improve the software after deployment.

2) Write 3 differences between Agile and Waterfall models.

Ans:

1. Agile is flexible; Waterfall is rigid.
2. Agile allows requirement changes; Waterfall does not easily allow changes.
3. Agile performs testing continuously; Waterfall tests after development is finished.

3) Name the 3 main Scrum roles and write one line about the responsibility of each.

Ans:

- **Product Owner** – Defines requirements and prioritizes the product backlog to maximize business value.
- **Scrum Master** – Facilitates the Scrum process, removes obstacles, and helps the team follow Scrum practices.
- **Development Team** – Designs, develops, tests, and delivers the product increment during each sprint.

4) What is a Sprint in Scrum? Mention its typical duration.

Ans: A Sprint is a fixed period during which the Scrum team works to complete a set of planned tasks and deliver a usable product increment.

- A Sprint usually lasts **1 to 4 weeks**, with **2 weeks** being a common duration.

5) What do these Jira board columns mean: To Do, In Progress, Done, Backlog?

Ans:

- 2) **Backlog** – List of all tasks, user stories, and issues that are planned but not yet started.
- 3) **To Do** – Tasks that are ready to be worked on but have not started yet.
- 4) **In Progress** – Tasks that are currently being worked on by the team.
- 5) **Done** – Tasks that have been completed and finished successfully.